

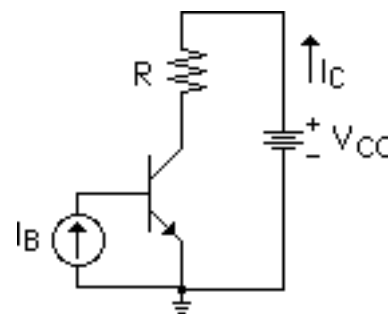
BJT Biasing

Objective

In general all electronic devices are nonlinear, and device operating characteristics can vary significantly over the range of parameters over which the device operates. The bipolar transistor, for example, has a 'normal' operating collector voltage range bounded by saturation for low voltages and collector junction breakdown for high voltages. Similarly the collector current is bounded by dissipation considerations on the one hand and cutoff on the other hand. In order to function properly the transistor must be biased properly, i.e., the steady-state operating voltages and currents must suit the purpose involved. Our primary concern here however is not to determine what an appropriate operating point is. That determination depends on a particular context of use and even so often involves a degree of judgment in choosing between conflicting preferences. Rather we suppose in general that an operating point is specified (somehow) and the task considered is how to go about establishing and maintaining that operating point. Where a specific context is needed for an illustration we assume usually that the transistor is to provide linear voltage amplification for a symmetrical signal, i.e., a signal with equal positive and negative excursions about a steady-state value.

BJT Amplifier

We start with an examination of a more or less general circuit to provide a broad introduction to a consideration of biasing. Thus consider the simplified bipolar transistor amplifier circuit drawn to the right. A current source in the base loop forward-biases the emitter junction and sets the base current to a fixed value I_{BQ} . Provided the voltage drop across the collector resistor is not too large the collector junction is reverse-biased, and the transistor is in its normal forward operating mode. The collector current is a function of the base current; $\Delta I_C \approx \beta \Delta I_B$. If then a small change is made in I_B there is a corresponding change in collector current, and a consequent change in the voltage drop across the resistor. The battery (DC) voltage V_{CC} provides a source of energy in the collector circuit. The battery provides each coulomb of charge carried by the collector current around the loop with the ability to do V_{CC} joules of work. Part of this work-doing ability provided, $I_C R$ joules, is expended in the collector resistor. The remainder, $V_{CC} - I_C R$ joules, is dissipated in the transistor. The rate of doing work, i.e., power, is determined by the collector current, i.e., the rate of charge transport. Hence by controlling the current the power provided by the battery and divided between the resistor and the transistor is controlled. (In this respect the power expended in the resistor should be interpreted as a general consumption of energy rather than simple ohmic dissipation, perhaps for example by a loudspeaker or a small fan motor.)



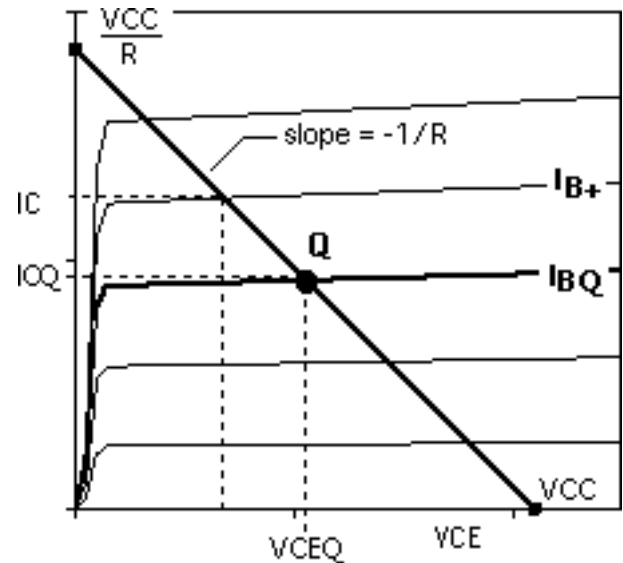
The transistor provides the current-control capability by acting as a current valve; a change in base current causes a corresponding amplified collector current change. The change in power expended in the collector resistor can be considerably greater than the power needed to cause the change in base current. The base current itself is much less than the collector current; small changes in base current readily cause collector current an order of magnitude or more larger. Moreover only a small emitter junction voltage change is needed to change base current considerably. The power that must be provided at the transistor base to effect a power change in the collector loop is therefore the product of a relatively small base current change and a similarly relatively small emitter junction voltage change. On the other hand not only is the collector current much larger than the base current but also the battery voltage ordinarily is much larger than the base voltage, allowing larger collector voltage changes.

To solve for the loop current one could write $V_{CE} = V_{CC} - I_C R$, a KVL loop equation, and substitute for V_{CE} from the volt-ampere relation of the transistor. It is convenient to illustrate the solution graphically, particularly so because the transistor volt-ampere relation is nonlinear. Thus the transistor collector characteristics are plotted (sans numerical values for simplicity), and superimposed on the plot is a graph

of the KVL loop equation. This latter equation plots as a line; a convenient way of doing this is by locating the axis intercepts as shown. Since the initial base current is I_{BQ} the operating point (solution) must lie somewhere along the emphasized collector characteristic curve; in other words the transistor volt-ampere relationship must be satisfied.

Concurrently the KVL expression must be satisfied; the operating point also must lie on the 'load' line.

Therefore the operating point must be the intersection of the two curves, i.e., the point Q (for 'quiescent').



Suppose now the base current is changed, say increased to a value I_{B+} . There is a consequent increase in collector current, an increase in the voltage drop across the resistor, and a decrease in the collector-emitter voltage. Locating the new operating point graphically is a matter of moving 'up' the load line to the intersection with the transistor characteristic identified by the base current I_{B+} .

Biasing the Transistor Amplifier

Converse to analyzing a specified amplifier circuit to determine the quiescent point is 'biasing' the transistor, i.e., arranging for a specified quiescent operating point. There are two aspects to this synthesis: first deciding where the quiescent point is to be located, and then locating it appropriately.

There is no unequivocal answer as to the proper quiescent point; it depends on what sort of performance the amplifier is to provide. For example suppose the amplifier is to provide a symmetrical voltage swing about the Q point. For a maximum symmetrical swing the Q point should be located at roughly $V_{CC}/2$ (neglecting the few tenths volt saturation voltage). For a lesser amplitude swing the Q point might be located lower down on the load line; this would involve lower collector currents and therefore lowered requirements on the power supply (here a battery). Indeed for certain common applications the Q point is located (roughly) at $(V_{CC}, 0)$; this provides for a maximum unipolar voltage swing.

For the present purpose we will not be overly concerned with a specific location of the Q point. By and large we concern ourselves here with just the means for establishing a given operating point, and not what the Q point values are. Actually more than just this. Transistor parameters have significant uncertainties; for example parameters are quite temperature sensitive, and in addition have large manufacturing tolerances. Not only must a specified Q point be implemented with uncertain circuit element parameters but also it must be maintained during environmental changes and kept the same from one copy of a circuit to the next despite device manufacturing tolerances.

Suppose, for example, the circuit analyzed above is subject to a temperature change of about 100°C ; in a temperate climate this corresponds roughly to the ability to function either on a hot summer day or on a cold winter day. The current amplification factor β varies typically by a factor of roughly two over this temperature range. Hence the collector current will differ by a factor of about two also, and this variation does not make for a particularly stable operating point providing consistent operation in both summer and winter. Besides this variation a production run of electronic equipment uses devices whose specifications vary significantly because of manufacturing tolerances; all the equipment nevertheless should function properly.

Biasing Against a Reference

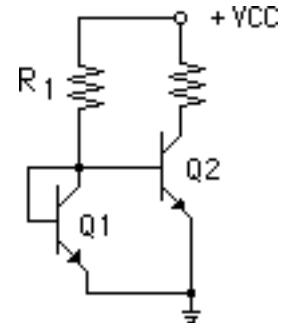
We assume that the transistor considered will operate in the normal forward operating range. Actually this is not always a design objective but it serves to illustrate concepts; exceptions will be illustrated in later work. Incidentally assuming this is the operating mode does not actually make it so; hence as a general

caveat always assure explicitly that a final design actually is consistent with the assumption made in determining the design.

To repeat, the general intent here is to take as given a specified operating point, and in addition some sort of constraint on its variability to perturbations, and to implement this intent.

There are (roughly) two kinds of biasing constraints to consider, either an absolute and a relative constraint. An absolute constraint fixes (say) a collector current range absolutely, e.g., requires $2\text{ma} \leq I_C \leq 2.1\text{ma}$. Note that 'absolute' does not mean 'constant'. A relative constraint, on the other hand, specifies how closely two currents (say) are to track each other even if both currents individually vary absolutely, e.g., $|I_{C1} - I_{C2}| \leq 0.1\text{ma}$. Both types of specifications can apply concurrently, e.g., one BJT current can be specified absolutely, and other BJT currents can reference that current in a relative specification.

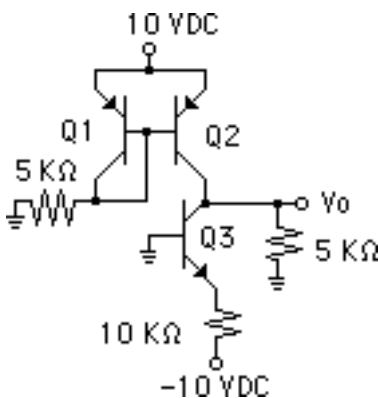
The simplified circuit to the right, in one variation or another, is commonly used effectively in integrated circuit design where circuit components closely matched in characteristics are a feature of the technology. The discussion following is primarily qualitative. Q_1 and Q_2 are matched components; Q_1 is operated as a 'diode-connected' transistor. (This is transistor and not diode operation; the collector-base junction is zero-biased, not forward-biased and the transistor is not saturated.) The intent is that the emitter currents of both transistors be the same (closely), and this intention is to be accomplished by having the same emitter junction voltage for the two matched devices.



Assuming a reasonably high value for β the base current of Q_2 will be much less than the emitter current of Q_1 , i.e., the current through R_1 is closely equal to the Q_1 emitter current. If the supply voltage V_{CC} is made significantly larger than the uncertainty in the emitter junction voltage (i.e., variations from a nominal 0.7 volt) then the Q_2 current will be determined to a high degree of precision by V_{CC} and R_1 . Here then is a case where the Q_1 current is constrained absolutely and the Q_2 current is constrained relative to the Q_1 current. Note that additional stages (within reason) similar to the Q_2 stage can be referenced to Q_1 .

Actually there is inherently a modest difference between the Q_1 and Q_2 currents due to the Early Effect, unlike Q_1 the collector junction voltage of Q_2 is not small. This current difference can be estimated from the Early voltage but ordinarily after circuit parameters have been selected ignoring the Early Effect a computer analysis using nonlinear transistor models can be used to make necessary adjustments.

A variation on the preceding illustration biasing, also not uncommon in integrated circuit designs, is shown to the left. It is used here in part as an example of a biasing circuit, and in part as an illustration of how much useful information a little understanding can provide essentially by inspection. For example it is not difficult to estimate V_O to be about 4.65 volts? Consider the matter further.



The current in the diode-connected transistor Q_1 is estimated to be $(10 - 0.7)/5 = 1.86\text{ma}$. This assumes the forward-biased emitter junction voltage drop of the diode-connected transistor is about 0.7 volt. If Q_1 and Q_2 are reasonably well-matched devices (they will be in an integrated circuit) their respective emitter currents will be about the same (both devices have the same emitter junction bias). Actually the Q_2 current will be somewhat larger because of the Early Effect. The Q_2 emitter current magnitude is essentially the Q_2 collector current magnitude as well, assuming that the β of the transistor is reasonably high. (The collector current is 98% of the emitter current for β as low as 50.) Again estimating an emitter junction forward-bias voltage of 0.7 volt, this time for Q_3 , estimate the Q_3 emitter current, and so also the Q_3 collector

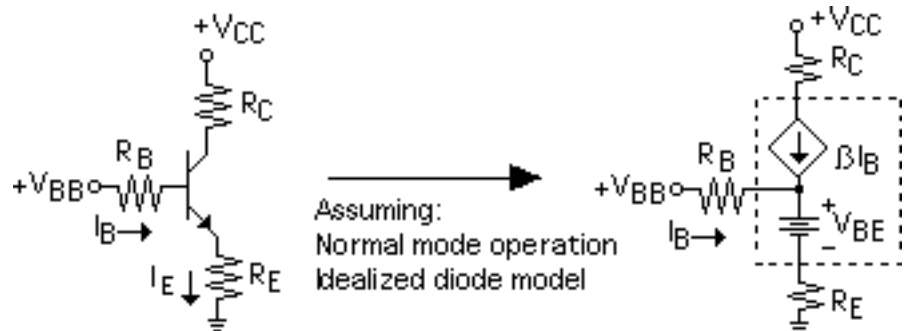
current, to be about 0.93 ma. The current through the 5k Ω load is the difference of the Q2 and Q3 currents, and so $V_o \approx 5 (1.86 - 0.93) = 4.65$ volts. Perhaps not exactly obvious by inspection, but nevertheless not difficult to estimate quickly.

Note that the biasing is not critically dependent on the transistor parameters, given a reasonably high β . Assume Q₁ and Q₂ are 2N3906 transistors, and Q₃ is a 2N3904. Use PSpice to compute the bias voltages and currents, and compare with the estimates (see Problem 2 below).

Emitter-Stabilized Biasing

Next we consider an 'absolute' biasing in which the range of variation of the collector current is prescribed; the basic circuit

was in fact considered earlier. The circuit as discussed here is shown to the right. It is the same as the circuit used earlier except that base current is obtained from a voltage source and, for reasons to be discussed, an emitter resistor is added. Although this biasing circuit is a simplified one it is not without subtlety. The objective is to bias the emitter current to a value $I_E \pm \Delta$ where Δ is a prescribed allowed variation.



A nonlinear calculation as a practical matter would have to be a computer numerical analysis. However an approximate analytic evaluation, even if not precise, can provide a much clearer appreciation of the role and influence of the various circuit components in biasing the transistor. For such a purpose the transistor is replaced, as an approximation, by an idealized PWL transistor model. Further, since normal forward operation is postulated the collector and emitter diodes respectively will be reverse- and forward-biased, and rather than clutter the diagram these diodes are 'drawn' as open- and short-circuits respectively. It is worth emphasizing here an earlier caveat; after circuit element values have been specified verify that assumed diode states are what the circuit voltages and current actually support; they might very well not be what you think they are.

First some formal algebra, and then an interpretation of the expressions obtained. The emitter current (in the PWL model) is the sum of the base and collector currents. The equation below is simply a loop equation (solved for I_E) written around the base-emitter loop, with the emitter current replacing the base current using the KCL expression $I_E = I_B(\beta+1)$.

$$I_E = \frac{V_{BB} - V_{BE}}{R_E + \frac{R_B}{\beta+1}}$$

Before leaping into calculations let us first consider this KVL expression qualitatively. Observe that the transistor properties enter into the equation in just two places, through V_{BE} and through β . Recall that V_{BE} varies with temperature; the voltage to sustain a given emitter current decreases as temperature rises by roughly 2 millivolt per $^{\circ}\text{C}$. Over a 100°C range that change is substantial, about 0.2 volt compared to the nominal 0.7 volt junction voltage drop. However it is not difficult to limit the effect of this change on the emitter current. To do so simply make the numerator as a whole 'large' compared to changes from the nominal value of V_{BE} (not to V_{BE} itself). A specific value for 'large' depends on a context, i.e., how much variation is acceptable in a given circumstance. The uncertainty in V_{BE} with temperature can, as noted, be estimated as something of the order of 0.2 volts over 100°C for silicon devices. If a factor of 10 is adequately 'large' make $V_{BB} > 2.5$ volts or so. Then the variation in emitter current due to the uncertain value associated with V_{BE} is less than $0.2/(2.5 - 0.7) \approx 11\%$.

The current amplification parameter β also is an uncertain transistor parameter; temperature variation, manufacturing tolerances and other dependencies can cause uncertainties from a nominal value by a factor of two or more. The value of β from one device to another of the same type can vary considerably because of manufacturing tolerances. The influence of β on the value of the emitter current can be reduced by making the term $R_E \gg R_B/(\beta+1)$ in the denominator of the expression. For a nominal minimum β of 99, and if $R_B < 10 R_E$ the term $R_B/(\beta+1)$ is no greater than $0.1R_E$.

Incidentally it might be asked why not make R_B zero, i.e., just make a direct connection and reduce the term involving β to zero. No reason really not to consider doing so, if biasing were the only consideration involved. When application of the device is considered, e.g., use as a signal amplifier, an argument can be made (later) for making R_B very large. Such conflicts are typical concerns to be resolved in electronic circuit design. In fact if you think about it using the amplifier for amplification implies that the emitter current will have to vary to some extent at least to track an input signal, so the current really should not be held precisely constant! In a latter note on amplifiers we consider these matters in more detail.

Illustrative Bias Calculation

Suppose (for the preceding circuit) an emitter bias current of 1 ± 0.1 milliamperes is to be provided for the circuit under discussion; specify appropriate values for V_{BB} , R_B and R_E . This problem is not one entirely susceptible to straightforward analysis, i.e., there are more variables than there are independent equations available to compute all parameters.

Since the bias design will (by design!) make the emitter current insensitive to transistor parameters we need not be overly concerned here about the selection of a particular device for this purpose. Moreover we will use an iterative procedure, i.e., work out a design, analyze the design (prospective component values being then known), and if the specifications are not met adjust the design appropriately. It is worth emphasizing yet again that simply asserting design specifications does not necessarily make those specifications realizable.

To make V_{BE} small compared with an estimated ± 0.1 volt variation for a nominal $\pm 50^\circ\text{C}$ variation about room temperature choose V_{BB} to be 3 volts. This is not a value that can be calculated; one could well choose other values for a trial design. What is important to note is that this choice, which can be adjusted later if desired, is not simply an arbitrary one. It is based on the consideration that the term $V_{BB} - V_{CE}$ should be 'large' compared to the ± 0.1 volt uncertainty in emitter junction voltage; here the ratio is about $0.1/(3-0.7) \approx 4\%$. Whether it should be 4%, or 1% or 10% is a matter for subjective evaluation of objectives in a specific context.

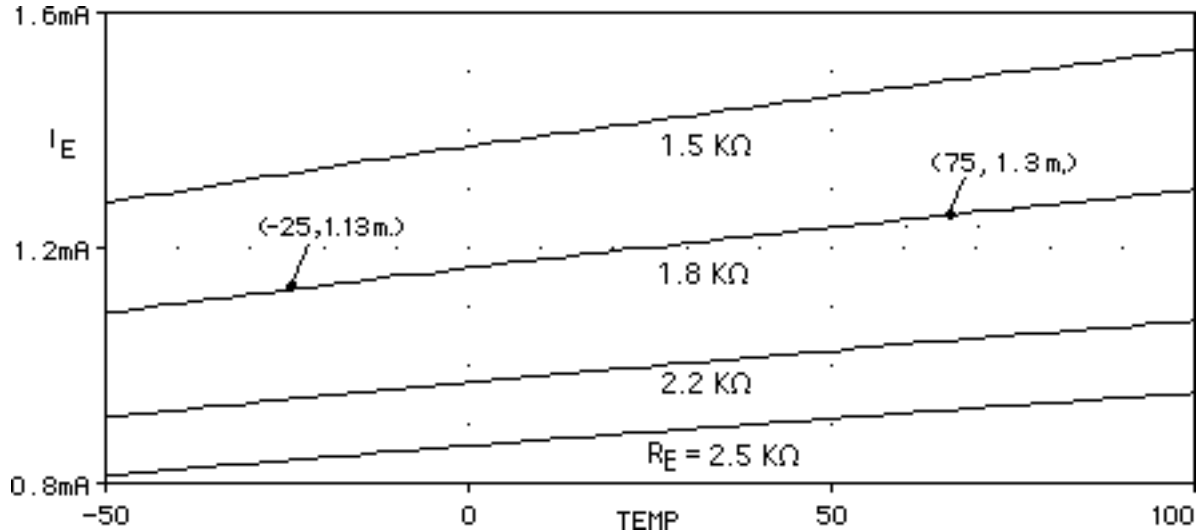
The voltage drop across R_B will be made small compared to that across R_E to reduce the influence of β , i.e., $I_E R_E \ll I_E R_B/(\beta+1)$. (Note that $I_B \approx I_E/(\beta+1)$). Hence the voltage drop across the emitter resistance will be about $3 - 0.7 = 2.3$ volts. For a 1 ma emitter current this would call for a 2.3 K Ω resistor, not a standard value. We can use either 2.2 K Ω or 2.5 K Ω in a standard 5% tolerance resistor. The smaller value means the emitter current will be somewhat higher than 1 ma, the larger value means a somewhat lower current. Choose a trial value of, say, 2.2 K Ω . Assume the transistor will have a minimum β of (say) 100; this is a matter of device selection.

Specify R_B so that the term involving β is less than (say) $0.1R_E$; $R_B \ll (\beta+1) R_E$ will do this. Assuming $\beta \approx 100$ or so choose $R_B = 22 \text{ K}\Omega$ (standard 5% value). Here again more of an educated choice rather than a calculation is involved; choosing $R_B = 0.1(\beta+1) R_E$ means allowing β less than a 10% influence on the value of the denominator.

Note that the collector resistance does not enter into these calculations directly. However if the collector junction is to be reverse-biased (unsaturated operation) the collector voltage must remain greater than about 3 volts. This will limit the maximum size of R_C usable for a given V_{CC} and collector current.

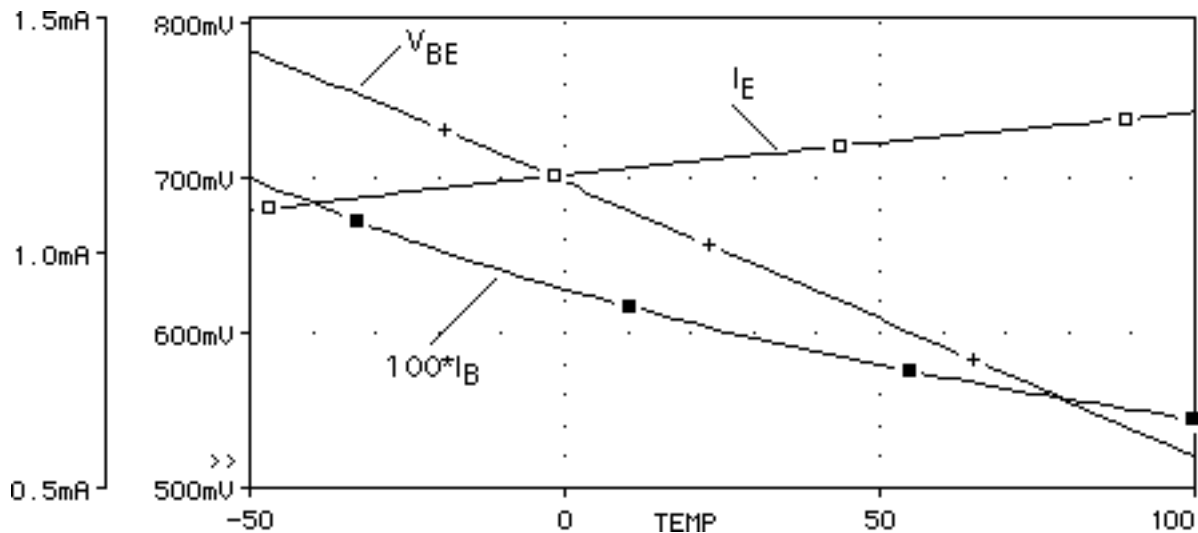
A PWL circuit analysis using the selected values calculates a nominal current of 0.95 ma. If β were to be larger than the minimum value of 100 assumed the current would increase to no more than 1.05 ma. The V_{BE} uncertainty corresponds to ± 0.04 ma or so.

Now that there is a nominal design there are several ways to adjust the design. For example a PSpice computation using a nonlinear 2N3904 transistor model provides the results plotted below. The emitter current is plotted vs. temperature for several choices of emitter resistance (standard 5% values); the base resistance is fixed at 22 K Ω . Computed current values are shown at -25°C and 75°C for one resistance value; the change in emitter current over this 100°C temperature range is 0.17 ma, with a nominal value at 27°C of about 1.2 ma.



Incidentally it is worth noting that being dogmatic about meeting the bias current specification can make the circuit design unnecessarily complicated. In most cases the bias current requirement could be changed, within limits, without requiring a major overall redesign. It may be easier and quite acceptable to change the specification to accommodate a different bias current. For example there might be advantage to using a 2.5 K Ω emitter resistance for a lower emitter current and a lesser absolute change in bias current with temperature. Such a question requires a context for a decision; the point here really is that this question and similar ones should not be set aside summarily.

The same bias circuit with R_E fixed at 1.8 K Ω provided the computed data plotted in the next figure. Note that V_{BE} changed about 2.5 mv over the 150°C temperature range. Note also that I_B changes quite a bit also, but the relative change in I_E is much smaller. The data indicate that β is increasing with temperature

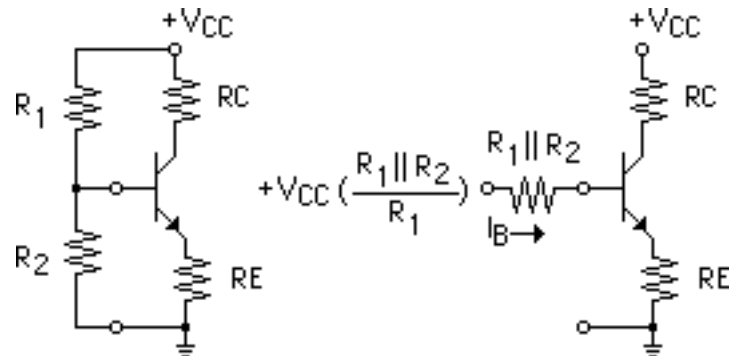


(as expected) but that the emitter resistor is acting to adjust the emitter junction voltage to reduce the change in emitter current from what would otherwise occur.

Single-Battery Biasing

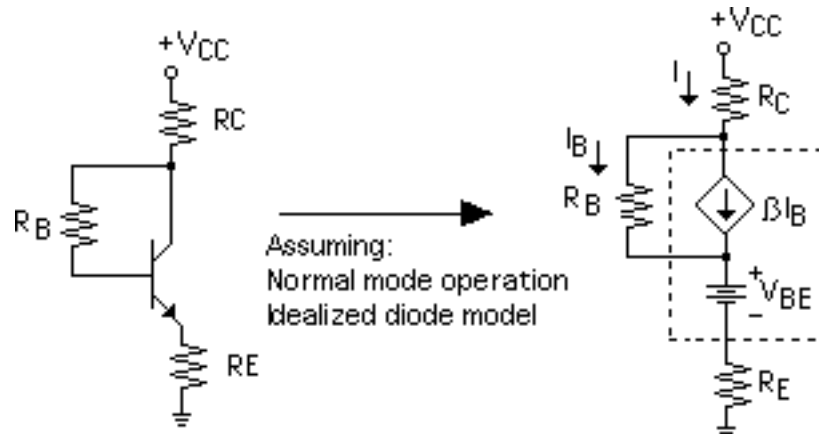
The circuit we have analyzed would not ordinarily be designed to use separate voltage supplies for the base and collector. Rather a single source would be used as illustrated in the figure below, left. Note that R_1 and R_2 do not form a resistive voltage divider; there is a base current so that the same current does not flow in the resistors. The analysis of this circuit can be reduced to that considered just above by determining the Thevenin equivalent circuit looking into R_2 . The Thevenin resistance is that calculated with all independent sources turned off (not removed), and is $R_1 \parallel R_2$. The Thevenin voltage is that appearing across the open-circuit terminals and is $V_{CC}(R_1 \parallel R_2)/R_1$. The transformed circuit is shown on the right of the figure, and with straightforward re-labeling is formally the same as that analyzed before. The influence of the circuit design is the introduction of the additional step of working backward to calculate R_1 and R_2 from V_{CC} , V_{BB} , and R_B . (Remember to adjust resistance specifications to standard values.

Incidentally the condition to minimize the influence of β is equivalent to making the current in $R_1 \gg I_B$ (so that base current variations cause small base voltage changes). Hence it is often a good first approximation in estimating the base voltage to neglect the base current and treat R_1 and R_2 as if they did in fact form a voltage divider. On the other hand accounting for the base current explicitly is not particularly difficult.



Collector-Stabilized Biasing

Emitter-stabilized biasing uses an emitter resistor to react to a change in emitter current and adjust the emitter junction bias so as to mitigate the change in emitter current. A complementary method is to use a base resistor to do a similar sort of thing. A circuit implementation is shown below, together with an idealized diode circuit model. (R_E is included simply to extend the discussion, i.e., the circuit has both emitter and collector stabilization.)



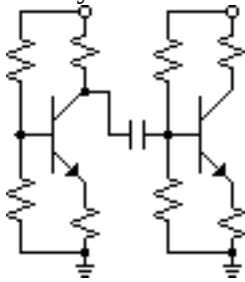
If the collector (or emitter) current changes from a design value because of some environmental change or transistor manufacturing tolerance the base current also changes. Applying KCL it follows that the current in R_C and the emitter current are equal, and both are equal to $(\beta+1) I_B$. Indeed (and here we include R_E)

$$I = \frac{V_{CC} - V_{BE}}{R_C + R_E + \frac{R_B}{\beta + 1}}$$

It is clear that R_C serves in the same way as R_E to offset the influence of the transistor β . One difficulty with collector stabilization, limiting its applicability, is that making $R_B/(\beta+1) \ll R_C+R_E$ also means the voltage drop across the transistor itself must be relatively small, limiting the allowable magnitude of a collector voltage change.

Capacitor-Coupled BJT Biasing

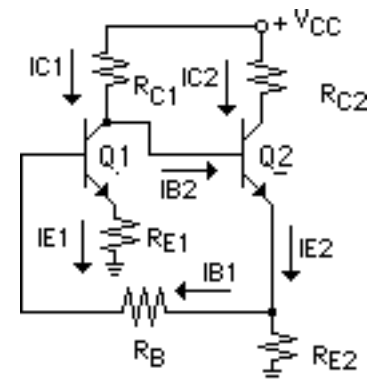
In many if not most instances a single amplifier stage is not adequate to the need; a multistage amplifier must be used. The two-stage circuit illustrated uses perhaps the simplest method of coupling a pair of stages, at least insofar as biasing is concerned. The capacitor, provided it has a small enough reactance at frequencies of interest passes a varying signal from the first to the second stage. DC voltages and currents are of course blocked, and the biasing calculations for one stage are independent of the other. Unfortunately the AC coupling inevitably degrades amplification for low enough signal frequencies. For some applications this is not acceptable, and in general there is a cost to providing an adequately large capacitance. In anticipation of this other representative transistor biasing pairs are considered.



Shunt-Series BJT Biasing

The two transistors in the biasing arrangement shown below left are 'direct coupled', i.e., there are no frequency-sensitive elements used for the biasing which might affect the frequency range over which the circuit could be used for amplification. As will be developed biasing stability is obtained for Q1 essentially independently of Q2; biasing stability for Q2 then is derived in part from that for Q1.

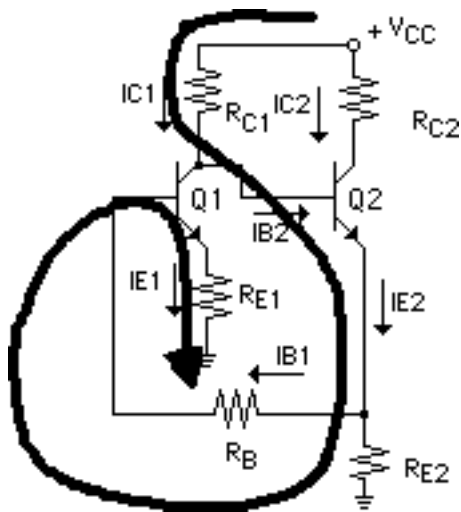
Even using idealized diode models for the transistors the amplifier circuit analysis still involves solving a pair of simultaneous equations; because of a seemingly general perversity of nature this is certain to involve complicated expressions which turn out to make numerically negligible contributions. Rather than endure this we might employ an iterative analysis procedure (which often provides a more than adequate approximate answer with a single iteration).



First make a reasonable simplifying assumption (not a random guess) about the circuit, and analyze the simplified circuit. Of course it will be necessary to use the results of the simplified analysis to validate the simplification, i.e., verify its consistency with the calculations. The underlying support for this process is that the circuit solution is unique. This sort of approach is generally applicable, although 'simplifying assumptions' should be determined separately for each case. There is also another kind of strong support for the use of iterative approximation that we describe later.

Consider the present circuit. A transistor used in an amplifier may be reasonably expected to have a reasonably high β . Hence the base current is virtually certain to be much smaller than the collector or emitter current. Suppose we will assume (always subject to later verification) that the Q2 base current is small compared to I_{C1} , and neglect it in writing a KCL equation at the Q1 collector-Q2 base. Note that it is misleading to suggest that the Q2 base current simply be set to zero; that can lead to conceptual problems. Rather simply neglect one term by comparison to another in a KCL equation at the node. Then it is clear how this assumption can be verified later; we determine the Q2 base current from the calculated Q2 emitter current, and compare it to the calculated Q1 collector current.

This base current assumption also clarifies the conceptual basis of this biasing configuration. The collector voltage of Q1 and the emitter voltage of Q2 differ only by the nearly constant Q2 emitter-junction voltage drop (large current changes involve exponentially smaller voltage changes). Hence except for a small offset voltage the Q1 base resistance is more or less connected electrically to the Q1 collector, just the arrangement described for collector stabilization. Since the collector current is stabilized so also is the Q1 collector voltage. Emitter stabilization of Q2 (the base voltage variation is constrained) completes the design as shown.



Analysis of the circuit (after element values are selected) is most easily done by writing a single equation around the contour shown in the accompanying figure. This contour takes advantage of neglecting the Q2 base current by using I_{C1} as the Q1 collector current. Then the Q1 base current is I_{C1}/β , and the emitter current is $(\beta+1) I_{C1}/\beta$.

The loop voltage equation is

$$V_{CC} = I_{C1}R_{C1} + V_{BE2} + \frac{I_{C1}}{\beta_1} R_B + V_{BE1} + \frac{\beta_1 + 1}{\beta_1} I_{C1}R_{E1}$$

and with some algebraic manipulation

$$I_{C1} = \frac{V_{CC} - V_{BE2} - V_{BE1}}{R_{C1} + \frac{\beta_1 + 1}{\beta_1} R_{E1} + \frac{R_B}{\beta_1}}$$

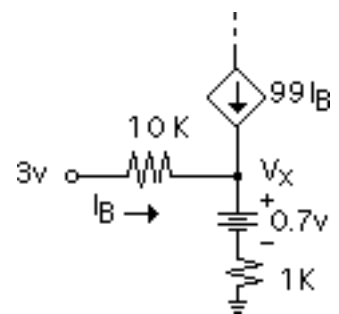
For most purposes it usually turns out to be unnecessary to refine the trial design calculation further. Keep in mind that the circuit is stabilized so that changes from the operating point are mitigated. A mathematical difference in the value of the Q2 base current from the exact value corresponds to a perturbation of the operating point. Changes in other currents and voltages caused by this perturbation are reduced by the stabilization, and therefore what is calculated is closer to the undisturbed values than otherwise might be.

Refining a calculation also is assisted by the built-in circuit stabilization. Use the calculated Q2 emitter current to estimate the base current. Then repeat the calculation, this time using the estimated base current value rather than simply neglecting the base current. Repeated iteration converges very quickly, particularly since the circuit design is likely to call for a high level of stability. An illustration of this follows.

The figure shows an idealized diode model of an emitter-stabilized amplifier operating in normal forward mode. Assume an initial guess is made for the value of I_B . Calculate the voltage $V_X = 3 - 10 I_B$ (I_B in ma). From this determine $I_E = V_X/0.7$ (I_E in ma). Finally estimate I_B from the calculated

n	$I_B(n)$	$V_X(n)$	$I_E(n)$
0	1	-7	-7.7
1	-0.77	10.7	10
2	0.1	2	1.3
3	0.13	1.7	1
4	.022	2.9	2.2
5	.0208	2.78	2.08
6	.02092	2.792	2.092
7	.02091	2.791	2.091
8	.02091		

I_E as $I_E/100$. Use this revised value for I_B in the next iteration. A tabulation of iterated calculations is shown to the right. The initial value for I_B was chosen as 1ma, clearly not the best estimate that might be made. A better first estimate, for example, would neglect the voltage drop across R_B , and determine (for this condition the $I_E \approx 2.3$ ma, and $I_B \approx .023$ ma). The closer the initial guess the faster the convergence. The extreme choice of $I_B \approx 1$ ma as the initial guess is made simply to show that convergence is rapid even for this assumption. Incidentally for this simple circuit an exact calculation is not difficult; it leads to $I_B = 0.02091$ ma.



Complementary Pair Biasing Illustration

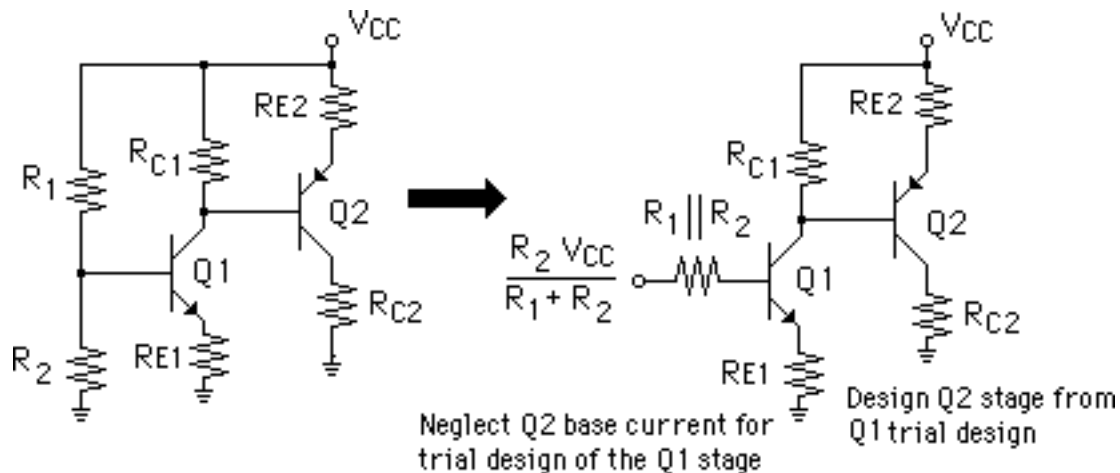
For many if not most purposes a design is best carried out as (roughly) an iterative two-step process. First an approximate, simplified model replaces a nonlinear device to enable a hand calculation to provide 'ballpark' design parameters; such simplified models can be used to provide approximate but nevertheless useful estimates of the relative importance of different parameters in obtaining the desired circuit

performance. (We did just this in using the PWL BJT model to estimate the emitter current, and the relative importance of the base and emitter resistors.)

Such a trial design result, possibly iterated to improve a calculation, then can be used to define the circuit parameters for which a numerical computation is made using nonlinear device models. From the numerical results one (in general) adjusts the trial design using the approximate relationships between parameters as a guide.

Here is an example of this process for a 'complementary biasing pair'. If NPN transistor stages simply are cascaded successive collector voltages are forced continually higher. This is because the collector voltage of one stage feeds the base of a successor stage, and the collector voltage of that successor stage must be increased accordingly to avoid saturation. Unfortunately in an amplifier successive stages would involve increasing signal amplitudes, and so a greater range of collector voltage variation. Thus the available range of collector voltage variation is reduced as the need for a larger range increases.

The NPN-PNP complementary pair arrangement avoids this by a natural 'level shifting' that occurs. The PNP stage should have its emitter (what would be the collector of a NPN stage) voltage increased to forward bias the emitter junction. On the other hand the PNP collector voltage should be lower than the base voltage to maintain the PNP collector reverse biased. Note that the lowered PNP collector could be used to drive a successor NPN stage, i.e., form a complimentary 'triple'.



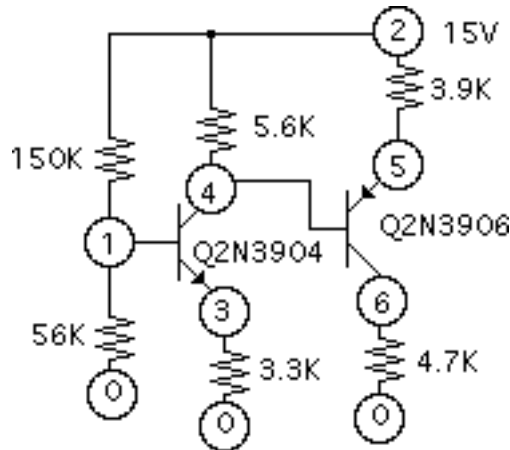
To simplify the analysis of the circuit somewhat a Thevenin transformation can be used to replace the input bias arrangement as shown; with a little practice this can be done without actually redrawing the figure. Formally we replace the transistor icons with a simplified PWL model, anticipating (subject to verification) that both devices are operating in the normal forward active mode. The problem then is reduced to a linear circuit analysis readily solvable. However even further useful simplification can be made to obtain a first trial design. Suppose we neglect (subject to later verification) the Q2 base current by comparison to the Q1 collector current; the transistor currents are likely to be of the same order of magnitude and β will be relatively large. This has the effect of 'separating' the two stages, making it fairly clear that the Q1 current is determined primarily by the base-emitter circuit of Q1. Hence design Q1 as if it were isolated, and then use the Q1 collector voltage and the well-defined emitter junction voltage drop to determine (approximately) the Q2 emitter voltage.

For example suppose V_{CC} is 15 volts, and the nominal Q1 and Q2 currents both are to be about 1 ma. Use a nominal $\beta = 120$ (since the effects of an uncertain β are to be constrained a precise value is not really needed). For a trial design set the Q1 base voltage to say ≈ 4 volts to constrain variations associated with the emitter junction voltage (simply an illustrative choice here). Choose $R_1 = 220K\Omega$ and $R_2 = 82K\Omega$ by the following argument. To limit the influence of the base current on the base voltage make the current through $R_1 \gg$ the base current. This means the base current can be neglected in estimating the R_1 - R_2 current as $15/(R_1+R_2)$ and setting this to be (say) $10 \times (1/120)$ estimate R_1+R_2 as $180K$, and then estimate R_2/R_1 as $4/11$. Use (rounding to standard 5% values) $R_1 = 150K$, $R_2 = 56K$.

For an emitter current of $\approx 1\text{mA}$ and with the emitter voltage at about 3.7 volts choose RE1 (standard value) to be $3.3\text{K}\Omega$. The minimum collector voltage to avoid saturation is approximately equal to the base voltage or about 4 volts. To allow the collector voltage to swing symmetrically between 15 volts (cutoff condition) and saturation (collector voltage = base voltage) choose RC1 to be $(15-4)/(2*1) \approx 5.6\text{K}$. Then to avoid saturating T2 limit the drop across RC2 to 4 volts, and so select RC2 = 3.9K .

Note: Primary concern here is for the biasing. There are still other considerations in an overall amplifier design, which are not considered as yet.

At this point circuit parameters have been assigned and, for example, a PWL model calculation can be made to improve the estimated circuit voltages and currents. Element values can be adjusted to modify a current or voltage as seems fit. Certainly check the calculations to see if there is consistency with numerical approximations and the assumed state of the diodes in the flag model. However this is left as an exercise. Instead the results of a PSpice analysis are shown next.



```
* Complementary Pair Biasing
R1 2 1 150K
R2 1 0 56K
Q1 3 1 4 Q2N3904
RC1 2 3 5.6K
RE1 4 0 3.3K
Q2 5 3 6 Q2N3906
RE2 2 6 3.9K
RC2 5 0 4.7K
VCC 2 0 DC 15
.LIB EVAL.LIB
.OP
.END
```

Node Voltages

V(1)= 3.8043V(2)= 5.0000
V(3)= 9.7353V(4)= 3.1420
V(5)= 5.4703V(6)= 0.4400

	Q1	Q2
MODEL	Q2N3904	Q2N3906
IB	6.70E-06	-5.31E-06
IC	9.45E-04	-1.16E-03
VBE	6.62E-01	-7.05E-01
VBC	-5.93E+00	4.27E+00
VCE	6.59E+00	-4.97E+00

Comparisons between computed and calculated PWL values are left as an exercise.

PROBLEMS

The problems below are **not** intended to be solved exactly algebraically. The circuits involve BJTs, and so fundamentally complex nonlinear terminal volt-ampere relations. To solve them as nonlinear problems is to substitute an inefficient hand calculation for what is readily done far faster, more accurately, and certainly more usefully numerically by a machine. There are circumstances however where a simplified inexact hand calculation to produce an approximate answer is quite useful. Hence the introduction of a (simplified) PWL model of the BJT, with the simplifications of the consequent linear (by segment) calculations. Even so a formal calculation even of a linear circuit can be arduous if a set of simultaneous equations, even a small number of equations, must be solved. A solution often involves numerous laborious expressions whose contribution to the overall result is largely negligible. It is certainly helpful if such terms can be largely suppressed by an approximate analysis. Learning to do this properly is a valuable engineering art as well as an aid to understanding. The problems following are intended to develop an appreciation of that art.

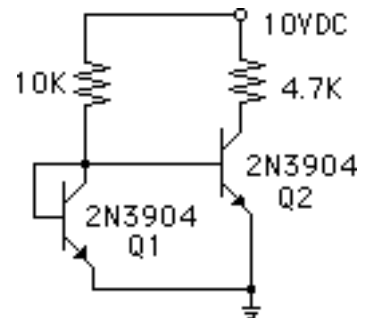
Some general guidelines (not rules) are:

- Reduce the nonlinear circuit to an adequate PWL model; the simplified idealized diode BJT model should be quite adequate for this purpose.
- Take advantage of the fact that the solution of a linear circuit is unique. Make reasonable approximations (on occasion simply an educated guess) for which there is some identifiable basis. Solve the circuit using the approximations, and then check the consistency of your approximations with the calculations. If your solution is reasonably consistent with your approximations, the solution is a reasonable approximation to the exact solution.
- For a BJT note that V_{BE} is ≈ 0.7 v, and β generally $\gg 1$ (i.e., base current $\ll$ collector current). Most bias circuits include a mechanism for stabilizing the biasing, i.e., limiting the effect of variations, particularly those associated with the transistor, from the design center. Since the circuit equations reflect the behavior of the circuit the effect of a 'variation' between a parameter value you assume and the true value of that parameter will be reduced by the stabilizing mechanism. The circuit actually reacts helpfully in converging to an acceptable degree of approximation.

For all problems here assume nominal 2N3904/2N3906 BJTs with $\beta \approx 120$ and $V_{BE} \approx 0.7$ V. A PSpice analysis including the .OP command provides computed circuit voltages and device currents.

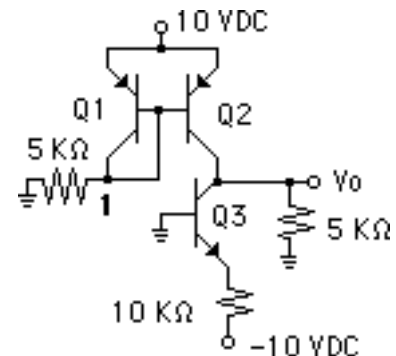
Problem 1)

Estimate the collector current for each transistor in the diode-connected transistor biasing circuit shown. Then compute and compare the collector currents over a temperature range -50°C to $+50^\circ\text{C}$.



Problem 2)

The current-mirror biasing circuit discussed earlier is redrawn to the right. Compute the circuit performance and compare with the estimates made earlier.

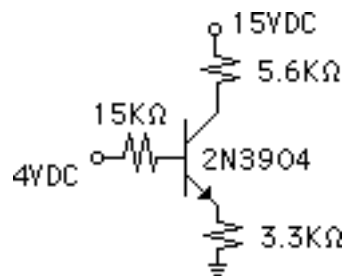


Problem 3)

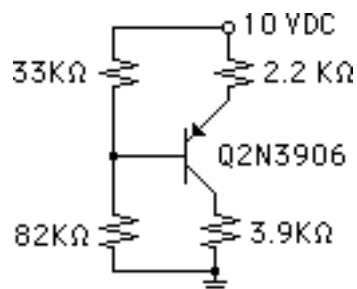
Given the circuit shown: estimate the collector current. Use the PWL model analysis that provided the relation

$$I_E = \frac{V_{BB} - V_{BE}}{R_E + \frac{R_B}{\beta + 1}}$$

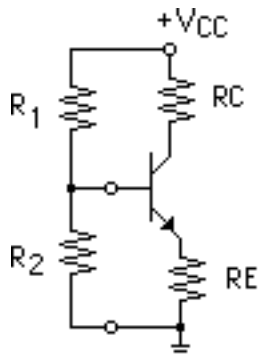
Is it a justifiable approximation to neglect the voltage drop across the base resistor compared to the 4 V source? Why?

**Problem 4)**

Estimate the collector current in the PNP bias circuit shown.

**Problem 5)**

a) Use the expression for the emitter current derived before for the circuit shown to derive the expression to the right for changes due to variation of V_{BE} and β . Interpret the expression in terms of the design guidelines discussed.



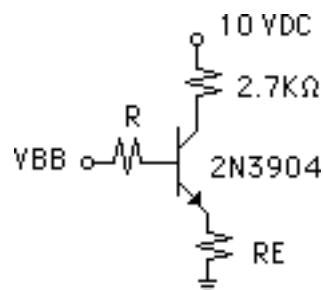
$$\frac{dI_E}{I_E} = \frac{d(V_S - V_{BE})}{V_S - V_{BE}} + \frac{1}{1 + (\beta + 1) \frac{R_E}{R_1 \parallel R_2}} \frac{d(\beta + 1)}{\beta + 1}$$

$$\text{where } V_S = \frac{R_2}{R_1 + R_2} V_{CC}$$

b) Evaluate the following argument. Suppose the collector current consists of a DC component and a sinusoidal component, i.e., $I_{CQ} + A \sin(\omega t)$. The average power provided by the source, $I_{CQ}V_{CC}$, is constant, and this is the same whether the AC signal is present or not. But the energy expended in the resistor increases since the average value of a sine-squared is not zero. Conservation of energy then requires the collector dissipation to be smaller with the AC signal present than in its absence. Thus the transistor operates cooler with a signal than without! Hence for safe operation with maximum AC signal the power rating for the transistor should be such that it is able to operate safely in the absence of an AC signal.

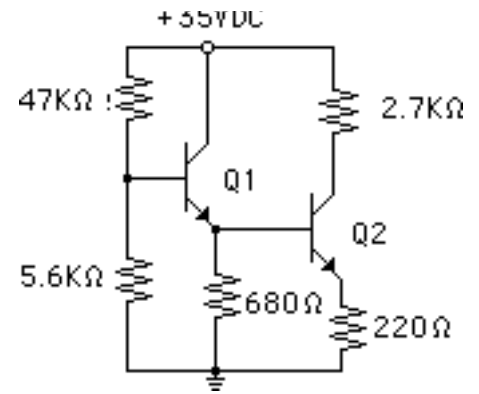
Problem 6)

Design the transistor circuit shown for a ± 0.1 ma variation about a nominal collector current of 1 ma. Assume a range of temperature variation of -50°C to $+50^\circ\text{C}$. Compute the collector current as a function of temperature for your design.

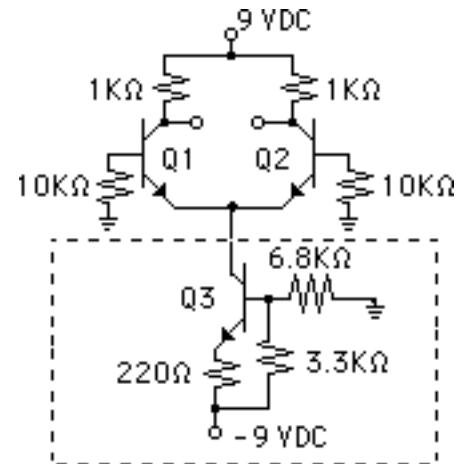


Problem 7)

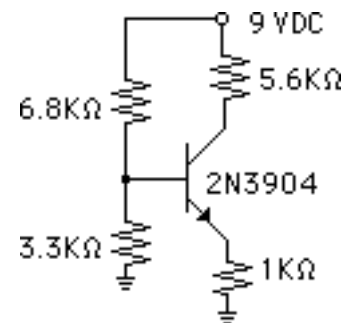
Assume (subject to subsequent verification) that the Q2 base current can be neglected compared to the Q1 emitter current. (Is this reasonable ?) Calculate the voltage at the collector of Q2. Assume 2N3904 devices.

**Problem 8**

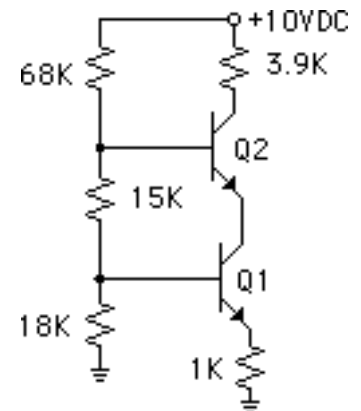
All the transistors (2N3904) in the symmetrical circuit shown are identical. Determine the collector voltage of Q2. Suggestion: Determine the collector current of Q3 first, and take advantage of the symmetry of the Q1, Q2 circuitry.

**Problem 9**

Calculate the emitter current in the circuit shown. Caution: Circuit designers who do not verify assumptions they make may be surprised.

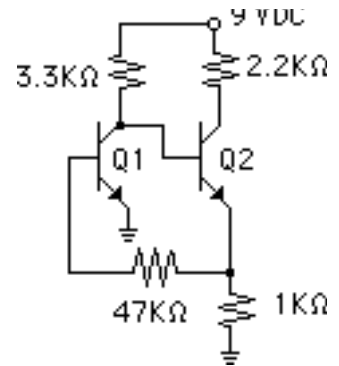
**Problem 10**

Anticipate (and later verify) that both transistor base currents are small compared to the current in the base biasing resistors. (That current will be roughly $10/(68+15+18) \approx 100 \mu\text{A}$. A conservative calculation neglecting the collector-emitter voltage drops indicates that the transistor collector current will be less than $10/(3.9+1) \approx 2 \text{ ma}$. With $\beta \approx 120$ the base currents then are estimated to be less than $20 \mu\text{A}$.) Estimate the collector current of Q2. Improve the collector current estimate by a recalculation using the base currents estimated from the first calculation, rather than neglecting them.

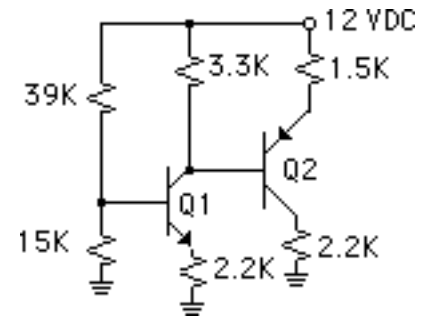


Problem 11

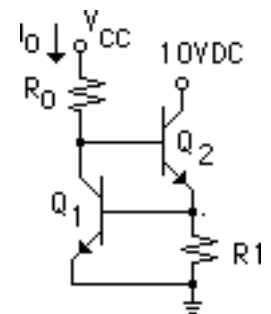
Calculate the collector voltage of Q2. Assume 2N3904 devices. Refer to the discussion above for simplifying suggestions.

**Problem 12**

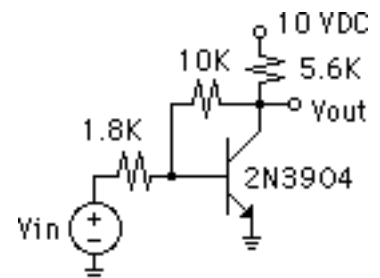
Determine the Q2 collector voltage for the complementary pair biasing configuration shown. Assume 2N3904 and 2N3906 devices.

**Problem 13**

A forward-biased junction operates over large current ranges with small voltage changes. The bias circuit shown takes advantage of this to reduce sensitivity to supply voltage variations. I_0 , assumed to have been made large compared to the base currents, is (approx.) $(V_{CC} - 2V_{BE})/R_0$. This current biases Q1 so as to provide a voltage V_{BE} across R_1 , so that the Q2 emitter current is approximately V_{BE}/R_1 . And V_{BE} is relatively insensitive to changes in V_{CC} . For $R_0 = 8.2K\Omega$, $R_1 = 470\Omega$, and $V_{CC} = 10$ volts estimate the Q2 collector current. Assume 2N3904 devices. Use PSpice to step V_{CC} from 0 to 10 volts, and plot $I_C(Q2)$ vs V_{CC} .



Problem 14) Calculate (approximately) the transfer characteristic V_{out} vs. V_{in} for the circuit shown, for $-1v \leq V_{in} \leq 1v$. Compare the calculated prediction against a computed plot.



Problem 15) Calculate (approximately) the transfer characteristic V_{out} vs. V_{in} for the circuit shown, for $0 \leq V_{in} \leq 10v$. Compare the calculated prediction against a computed plot.

